

HW2

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Note: We use $\subset \equiv \subseteq$ and $\supset \equiv \supseteq$ for the purposes of set relations. This note will be dropped henceforth in future problemsets.

Q1: A.29

Let $\mathbb{S} = C[0, 1]$ and $d_p(f, g) \equiv \left(\int_0^1 |f(t) - g(t)|^p dt \right)^{\frac{1}{p}}$ for $1 \leq p < \infty$ and $d_\infty(f, g) = \sup\{|f(t) - g(t)| : t \in [0, 1]\}$.

Note: May use the inequality:

$$|a + b|^p \leq 2^p (|a|^p + |b|^p)$$

(a)

Let $f(x) \equiv 1$. Let $f_n(t) \equiv 1$ for $0 \leq t \leq 1 - \frac{1}{n}$, and $f_n(t) = n(1 - t)$ for $1 - \frac{1}{n} \leq t \leq 1$. Show that $d_p(f_n, f) \rightarrow 0$ for $1 \leq p < \infty$ but $d_\infty(f_n, f) \not\rightarrow 0$.

Answer

Proof.

Case (1): $1 \leq p < \infty$. On $[0, 1 - 1/n]$, $f_n(t) = 1 = f(t)$, so $|f_n(t) - f(t)|^p = 0$ in that interval.

And on $[1 - 1/n, 1]$, $f_n(t) - f(t) = n(1 - t) - 1$.

It then follows that

$$d_p(f_n, f)^p = \int_0^1 |f_n(t) - f(t)|^p dt = \int_{1-\frac{1}{n}}^1 |n(1 - t) - 1|^p dt.$$

This can be evaluated; to do so, substitute $u = 1 - t$ and $du = -dt$ ($t = 1 - 1/n \Rightarrow u = 1/n$, $t = 1 \Rightarrow u = 0$):

$$d_p(f_n, f)^p = \int_0^{1/n} |nu - 1|^p du.$$

Utilize substitution again, this time with $w = nu - 1$ and $dw = n du$ ($u = 0 \Rightarrow w = -1$, $u = 1/n \Rightarrow w = 0$):

$$d_p(f_n, f)^p = \frac{1}{n} \int_{-1}^0 |w|^p dw = \frac{1}{n} \int_0^1 w^p dw = \frac{1}{n(p+1)}.$$

Since p is fixed, $\frac{1}{n(p+1)} \rightarrow 0$ as $n \rightarrow \infty$, so $d_p(f_n, f)^p \rightarrow 0$, and consequently $d_p(f_n, f) \rightarrow 0$, for every $1 \leq p < \infty$.

Case (2): $p = \infty$. On $[1 - 1/n, 1]$, $n(1 - t)$ is a strictly decreasing function of t , taking the value $n(1 - (1 - 1/n)) = 1$ at $t = 1 - 1/n$ and the value 0 at $t = 1$.

So $|n(1 - t) - 1|$ increases monotonically from 0 to 1 on this interval, attaining its maximum of 1 at $t = 1$, i.e.:

$$|f_n(1) - f(1)| = |n(1 - 1) - 1| = |0 - 1| = 1.$$

Since the difference is 0 on $[0, 1 - 1/n]$ and its supremum on $[1 - 1/n, 1]$ is 1 (attained at $t = 1$), it follows that for all $n \geq 1$

$$d_\infty(f_n, f) = \sup_{t \in [0, 1]} |f_n(t) - f(t)| = 1.$$

Hence $\lim_{n \rightarrow \infty} d_\infty(f_n, f) = 1 \neq 0$, and we conclude that $d_\infty(f_n, f) \not\rightarrow 0$.

(b)

Fix $f \in C[0, 1]$. Let $g_n(t) = f(t)$, $0 \leq t \leq 1 - \frac{1}{n}$, and $g_n(t) = f(1 - \frac{1}{n}) + (f(1) - f(1 - \frac{1}{n}))n(t + \frac{1}{n} - 1)$, $1 - \frac{1}{n} \leq t \leq 1$.

Show that $d_p(g_n, f) \rightarrow 0$ for all $1 \leq p \leq \infty$.

Answer

Sketch: For finite p , the Note plus boundedness of f gives $d_p(g_n, f)^p = O(1/n) \rightarrow 0$. For $p = \infty$, uniform continuity of f is sufficient to conclude.

Proof.

Fix $f \in C[0, 1]$. Since f is continuous on the closed and bounded interval $[0, 1]$, Proposition A.2.5(ii)(a) (continuous functions on a closed and bounded interval are bounded) gives $\sup\{|f(t)| : t \in [0, 1]\} < \infty$; fix $M > 0$ such that $|f(t)| \leq M$ for all $t \in [0, 1]$.

For $t \in [1 - 1/n, 1]$, let $s \equiv n(t + 1/n - 1) \in [0, 1]$ (the linear reparametrization of $[1 - 1/n, 1]$ onto $[0, 1]$).

By definition $g_n(t) = (1 - s)f(1 - 1/n) + sf(1)$, so

$$g_n(t) - f(t) = (1 - s)(f(1 - \frac{1}{n}) - f(t)) + s(f(1) - f(t)). \quad (1)$$

(Note: $(1 - s)f(1 - \frac{1}{n}) + sf(1) - (1 - s)f(t) - sf(t) = g_n(t) - f(t)$, using $(1 - s) + s = 1$.)

On $[0, 1 - 1/n]$, $g_n(t) = f(t)$, so the difference is 0 there.

Case (1): $1 \leq p < \infty$. By (1) and the Note, $|a + b|^p \leq 2^p(|a|^p + |b|^p)$, with $a = (1 - s)(f(1 - \frac{1}{n}) - f(t))$ and $b = s(f(1) - f(t))$:

$$|g_n(t) - f(t)|^p \leq 2^p \left[(1 - s)^p |f(1 - \frac{1}{n}) - f(t)|^p + s^p |f(1) - f(t)|^p \right].$$

Since $s \in [0, 1]$, $(1 - s)^p \leq 1$ and $s^p \leq 1$, so

$$|g_n(t) - f(t)|^p \leq 2^p \left[|f(1 - \frac{1}{n}) - f(t)|^p + |f(1) - f(t)|^p \right].$$

Since $|f(1 - \frac{1}{n})|, |f(t)|, |f(1)| \leq M$, both $|f(1 - \frac{1}{n}) - f(t)| \leq 2M$ and $|f(1) - f(t)| \leq 2M$, giving

$$|g_n(t) - f(t)|^p \leq 2^p [(2M)^p + (2M)^p] = 2^{p+1} (2M)^p$$

for all $t \in [1 - 1/n, 1]$.

Hence

$$d_p(g_n, f)^p = \int_0^1 |g_n(t) - f(t)|^p dt = \int_{1-\frac{1}{n}}^1 |g_n(t) - f(t)|^p dt \leq 2^{p+1} (2M)^p \cdot \frac{1}{n}.$$

Since p is fixed, $2^{p+1} (2M)^p$ is a constant, so $d_p(g_n, f)^p \rightarrow 0$ as $n \rightarrow \infty$, hence $d_p(g_n, f) \rightarrow 0$.

Case (2): $p = \infty$. Since f is continuous on the closed and bounded interval $[0, 1]$, Proposition A.2.5(ii)(c) gives that f is uniformly continuous on $[0, 1]$: $\forall \varepsilon > 0, \exists \delta > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$ for all $x, y \in [0, 1]$.

Let $\varepsilon > 0$ be given, and choose $N \in \mathbb{N}$ with $1/N < \delta$. Let $n \geq N$ and $t \in [1 - 1/n, 1]$ be given. Since $|t - (1 - \frac{1}{n})| \leq \frac{1}{n} \leq \frac{1}{N} < \delta$ and $|t - 1| \leq \frac{1}{n} \leq \frac{1}{N} < \delta$,

$$|f(1 - \frac{1}{n}) - f(t)| < \varepsilon \quad \text{and} \quad |f(1) - f(t)| < \varepsilon.$$

By (1) and the triangle inequality (the Note provided by Dr. Ghosh only covers finite p),

$$|g_n(t) - f(t)| \leq (1 - s)|f(1 - \frac{1}{n}) - f(t)| + s|f(1) - f(t)| < (1 - s)\varepsilon + s\varepsilon = \varepsilon$$

for all $t \in [1 - 1/n, 1]$. Combined with $g_n(t) - f(t) = 0$ on $[0, 1 - 1/n]$,

$$d_\infty(g_n, f) = \sup_{t \in [0, 1]} |g_n(t) - f(t)| \leq \varepsilon$$

for all $n \geq N$. Since $\varepsilon > 0$ was arbitrary, $d_\infty(g_n, f) \rightarrow 0$.

Combining both cases, $d_p(g_n, f) \rightarrow 0$ for all $1 \leq p \leq \infty$. //

Q2: A.34

(Part 1:) Show that unions of open sets are open and (Part 2:) intersection of any two open sets is open.
(Part 3:) Give an example to show that the intersection of an infinite number of open sets need not be open.

Answer

Proof.

Part 1: Unions of open sets are open.

Let $(\mathbb{S}, d)$ be a metric space, I an index set, and $\{O_\alpha\}_{\alpha \in I}$ a family of open sets on $\mathbb{S}$.

Let $x \in \bigcup_{\alpha \in I} O_\alpha$ be given. Then $\exists \alpha_0 \in I$ such that $x \in O_{\alpha_0}$.

Since O_{α_0} is open, $\exists \delta > 0$ such that for all $y \in \mathbb{S}$, $d(x, y) < \delta \Rightarrow y \in O_{\alpha_0}$.

Since $O_{\alpha_0} \subset \bigcup_{\alpha \in I} O_\alpha$, this gives $d(x, y) < \delta \Rightarrow y \in \bigcup_{\alpha \in I} O_\alpha$.

Since $x \in \bigcup_{\alpha \in I} O_\alpha$ was arbitrary, $\bigcup_{\alpha \in I} O_\alpha$ is open.

Part 2: The intersection of any two open sets is open.

Let A, B be open sets on $(\mathbb{S}, d)$. Let $x \in A \cap B$ be given, i.e., $x \in A$ and $x \in B$.

Since A is open, $\exists \delta_1 > 0$ such that for all $y \in \mathbb{S}$, $d(x, y) < \delta_1 \implies y \in A$.

Since B is open, $\exists \delta_2 > 0$ such that for all $y \in \mathbb{S}$, $d(x, y) < \delta_2 \implies y \in B$.

Let $\delta \equiv \min(\delta_1, \delta_2) > 0$.

Then for all $y \in \mathbb{S}$ with $d(x, y) < \delta$, both $d(x, y) < \delta_1$ and $d(x, y) < \delta_2$ hold, so $y \in A$ and $y \in B$, i.e., $y \in A \cap B$. Since $x \in A \cap B$ was arbitrary, $A \cap B$ is open.

Part 3: The intersection of infinitely many open sets need not be open.

Let $A_n \equiv (-1/n, 1/n)$ for each $n \geq 1$, an open subset of $(\mathbb{R}, |\cdot|)$. We claim $\bigcap_{n \geq 1} A_n = \{0\}$.

($\supset$) Since $0 \in (-1/n, 1/n)$ for every $n \geq 1$, $0 \in \bigcap_{n \geq 1} A_n$.

($\subset$) Let $x \in \bigcap_{n \geq 1} A_n$ be given, and suppose for contradiction $x \neq 0$.

By the Axiom of Archimedes (Proposition A.2.1: for all $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > y$), applied to $y = 1/|x|$, $\exists N \in \mathbb{N}$ such that $N > 1/|x|$, i.e., $1/N < |x|$, so $x \notin (-1/N, 1/N) = A_N$, contradicting $x \in \bigcap_{n \geq 1} A_n \subset A_N$.

Combining both containments, $\bigcap_{n \geq 1} A_n = \{0\}$.

Finally, $\{0\}$ is not open: for any $\delta > 0$, the point $y \equiv \delta/2$ satisfies $d(0, y) = \delta/2 < \delta$, yet $y \neq 0$, so $y \notin \{0\}$.

Thus no δ -ball centered at 0 is contained in $\{0\}$, so $\{0\}$ is not open, and we conclude that $\bigcap_{n \geq 1} A_n = \{0\}$ is a countable (infinite) intersection of open sets that is itself not open.//

Q3: A.36

Let $f_n(x) = x^n$ and $g(x) \equiv 0$ on $\mathbb{R}$. Then $\{f_n\}_{n \geq 1}$ converges pointwise to g on $(-1, 1)$, uniformly on $[a, b]$ for $-1 < a < b < 1$, but not uniformly on $(0, 1)$.

Answer

Proof.

(a) Pointwise convergence on $(-1, 1)$.

Let $x \in (-1, 1)$ be given. If $x = 0$, then $f_n(x) = 0 = g(x)$ for all $n \geq 1$, so convergence is immediate. If $x \neq 0$, then $|x| \in (0, 1)$, so $\log |x| < 0$. Let $\varepsilon > 0$ be given. If $\varepsilon \geq 1$, take $N \equiv 1$, since $|x|^n < 1 \leq \varepsilon$ for all $n \geq 1$ trivially. If $0 < \varepsilon < 1$, set

$$N \equiv \left\lceil \frac{\log \varepsilon}{\log |x|} \right\rceil + 1 \in \mathbb{N}.$$

For $n \geq N$, $n > \log \varepsilon / \log |x|$; since $\log |x| < 0$, multiplying both sides by $\log |x|$ reverses the inequality, giving $n \log |x| < \log \varepsilon$, i.e. $|x|^n < \varepsilon$. Thus $|f_n(x) - g(x)| = |x|^n < \varepsilon$ for all $n \geq N$.

In either case $\lim_{n \rightarrow \infty} f_n(x) = 0 = g(x)$. Since $x \in (-1, 1)$ was arbitrary, $\{f_n\}_{n \geq 1}$ converges pointwise to g on $(-1, 1)$.

(b) Uniform convergence on $[a, b]$ for $-1 < a < b < 1$.

Let $-1 < a < b < 1$ be given, and set $M \equiv \max(|a|, |b|)$. Since $a \neq b$, not both a, b equal 0, so $M \in (0, 1)$. For every $x \in [a, b]$, $|x| \leq M < 1$.

Similar to part (a) given above, let $\varepsilon > 0$ be given. If $\varepsilon \geq 1$, take $N \equiv 1$, since $|x|^n < 1 \leq \varepsilon$ for all $n \geq 1$ trivially. If $0 < \varepsilon < 1$, set

$$N \equiv \left\lceil \frac{\log \varepsilon}{\log |x|} \right\rceil + 1 \in \mathbb{N}.$$

By the same argument as in (a) (with M in place of $|x|$), $n \geq N_\varepsilon \implies M^n < \varepsilon$. Since $t \mapsto t^n$ is non-decreasing on $[0, \infty)$ and $0 \leq |x| \leq M$, $|x|^n \leq M^n$ for all $x \in [a, b]$. Thus, for all $x \in [a, b]$ and $n \geq N_\varepsilon$,

$$|f_n(x) - g(x)| = |x|^n \leq M^n < \varepsilon.$$

Since N_ε depends only on ε (via a, b) and not on x , $\{f_n\}_{n \geq 1}$ converges uniformly to g on $[a, b]$.

(c) Non-uniform convergence on $(0, 1)$.

We show the negation of uniform convergence holds: $\exists \varepsilon_0 > 0$ such that $\forall N \in \mathbb{N}$, $\exists n \geq N$ and $x \in (0, 1)$ with $|f_n(x) - g(x)| \geq \varepsilon_0$.

Let $\varepsilon_0 \equiv 1/2$. Let $N \in \mathbb{N}$ be given, and set $n \equiv N$ and $x_N \equiv (1/2)^{1/N}$; since $1/2 \in (0, 1)$, $x_N \in (0, 1)$ as well. Then

$$|f_n(x_N) - g(x_N)| = |x_N^N - 0| = \left(\left(\frac{1}{2}\right)^{1/N}\right)^N = \frac{1}{2} = \varepsilon_0 \geq \varepsilon_0.$$

Since $N \in \mathbb{N}$ was arbitrary, no N makes $|f_n(x) - g(x)| < \varepsilon_0$ hold for all $n \geq N$ and all $x \in (0, 1)$ simultaneously. Hence $\{f_n\}_{n \geq 1}$ does not converge uniformly to g on $(0, 1)$. //

Q4: 1.2

Let Ω be a finite set, i.e., the number of elements in Ω is finite. Let $\mathcal{F} \subset \mathcal{P}(\Omega)$ be an algebra. Show that $\mathcal{F}$ is a σ -algebra.

Answer

Sketch: By Proposition 1.1.1, it suffices to show $\mathcal{F}$ is closed under monotone unions to show that $\mathcal{F}$ is a σ -algebra. Since Ω is finite, an increasing sequence in $\mathcal{F}$ reduces the union to a finite union already in $\mathcal{F}$.

Proof. Since Ω is finite, write $m \equiv |\Omega| < \infty$; since $\mathcal{F} \subset \mathcal{P}(\Omega)$, every $A \in \mathcal{F}$ satisfies $A \subset \Omega$ and hence $|A| \leq m$.

Let $\{A_n\}_{n \geq 1} \subset \mathcal{F}$ be given with $A_n \subset A_{n+1}$ for all $n \geq 1$. Since $A_n \subset A_{n+1}$, $|A_n| \leq |A_{n+1}|$ for all n , so $\{|A_n|\}_{n \geq 1}$ is a non-decreasing sequence in $\{0, 1, \dots, m\}$. A non-decreasing sequence of integers bounded above by m can increase at most m times, so there exists $N \in \mathbb{N}$ such that

$$|A_n| = |A_N| \quad \text{for all } n \geq N.$$

Fix $n \geq N$. By monotonicity, $A_N \subset A_n$; combined with $|A_N| = |A_n| < \infty$, this gives $A_N = A_n$ (a finite set contained in a finite set of equal cardinality must coincide with it). Thus $A_n = A_N$ for all $n \geq N$.

Consequently,

$$\bigcup_{n \geq 1} A_n = \bigcup_{n=1}^N A_n,$$

since for $n \leq N$, $A_n \subset A_N$ by monotonicity contributes nothing beyond A_N , and for $n > N$, $A_n = A_N$ contributes nothing new. As $\{A_1, \dots, A_N\} \subset \mathcal{F}$ is a finite collection and $\mathcal{F}$ is an algebra (Definition 1.1.1(c), closure under finite unions),

$\bigcup_{n=1}^N A_n \in \mathcal{F}$. Hence

$$\bigcup_{n \geq 1} A_n = A_N \in \mathcal{F}.$$

Since $\{A_n\}_{n \geq 1} \subset \mathcal{F}$ with $A_n \subset A_{n+1}$ was arbitrary, $\mathcal{F}$ is closed under monotone unions. As $\mathcal{F}$ is an algebra by hypothesis and is closed under monotone unions, Proposition 1.1.1 (which states that $\mathcal{F} \subset \mathcal{P}(\Omega)$ is a σ -algebra if and only if $\mathcal{F}$ is an algebra satisfying $A_n \in \mathcal{F}$, $A_n \subset A_{n+1}$ for all $n \implies \bigcup_{n \geq 1} A_n \in \mathcal{F}$) gives that $\mathcal{F}$ is a σ -algebra (Definition 1.1.2). //

Q5: 1.4

Let $\{\mathcal{F}_\theta : \theta \in \Theta\}$ be a family of σ -algebras on Ω . Show that $\mathcal{G} \equiv \bigcap_{\theta \in \Theta} \mathcal{F}_\theta$ is also a σ -algebra.

Answer

Sketch: It suffices to verify (a), (b), (d) (Definition 1.1.1(a)-(b), Definition 1.1.2(d)); these force (c) (Definition 1.1.1(c)) automatically.

Proof.

(a) Since each $\mathcal{F}_\theta$ is a σ -algebra, $\Omega \in \mathcal{F}_\theta$ for all $\theta \in \Theta$ (Definition 1.1.1(a)). Hence $\Omega \in \bigcap_{\theta \in \Theta} \mathcal{F}_\theta = \mathcal{G}$.

(b) Let $A \in \mathcal{G}$ be given. Then $A \in \mathcal{F}_\theta$ for every $\theta \in \Theta$. Since each $\mathcal{F}_\theta$ is a σ -algebra, $A^c \in \mathcal{F}_\theta$ for every $\theta \in \Theta$ (Definition 1.1.1(b)). Hence $A^c \in \bigcap_{\theta \in \Theta} \mathcal{F}_\theta = \mathcal{G}$.

(d) Let $\{A_n\}_{n \geq 1} \subset \mathcal{G}$ be given. Then for every $\theta \in \Theta$, $A_n \in \mathcal{F}_\theta$ for all $n \geq 1$. Since each $\mathcal{F}_\theta$ is a σ -algebra, $\bigcup_{n \geq 1} A_n \in \mathcal{F}_\theta$ for every $\theta \in \Theta$ (Definition 1.1.2(d)). Hence $\bigcup_{n \geq 1} A_n \in \bigcap_{\theta \in \Theta} \mathcal{F}_\theta = \mathcal{G}$.

Since $\emptyset = \Omega^c \in \mathcal{G}$ by (a)-(b), (a), (b), (d) also give closure under pairwise unions (Definition 1.1.1(c)): for $A, B \in \mathcal{G}$, $A \cup B = A \cup B \cup \emptyset \cup \emptyset \cup \dots \in \mathcal{G}$ by (d). Hence $\mathcal{G}$ satisfies Definition 1.1.1 and Definition 1.1.2, so $\mathcal{G}$ is a σ -algebra. //

Q6: 1.6

Let Ω be a nonempty set and let $\mathcal{A} \equiv \{A_i : i \in \mathbb{N}\}$ be a partition of Ω , i.e., $A_i \cap A_j = \emptyset$ for all $i \neq j$ and $\bigcup_{i \geq 1} A_i = \Omega$. Let $\mathcal{F} = \{\bigcup_{i \in J} A_i : J \subset \mathbb{N}\}$ where, for $J = \emptyset$, $\bigcup_{i \in J} A_i \equiv \emptyset$. Show that $\mathcal{F}$ is a σ -algebra.

Answer

Sketch: As in Q5 (1.4), it suffices to verify (a), (b), (d) (Definition 1.1.1(a)-(b), Definition 1.1.2(d)); these force (c) (Definition 1.1.1(c)) automatically.

Proof.

(a) Since $\bigcup_{n \geq 1} A_n = \Omega$, taking $J = \mathbb{N} \subset \mathbb{N}$ gives $\Omega = \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{F}$.

(b) Let $B \in \mathcal{F}$ be given. Then $B = \bigcup_{i \in J} A_i$ for some $J \subset \mathbb{N}$. Let $J^c \equiv \mathbb{N} \setminus J$. We claim $B^c = \bigcup_{i \in J^c} A_i$.

Since $\{A_i : i \in \mathbb{N}\}$ is a partition, $A_i \cap A_j = \emptyset$ for all $i \neq j$; in particular, for any $i \in J$ and $j \in J^c$, $i \neq j$, so $A_i \cap A_j = \emptyset$.

Hence

$$\left(\bigcup_{i \in J} A_i \right) \cap \left(\bigcup_{j \in J^c} A_j \right) = \bigcup_{i \in J, j \in J^c} (A_i \cap A_j) = \emptyset.$$

Also, since $J \cup J^c = \mathbb{N}$,

$$\left(\bigcup_{i \in J} A_i \right) \cup \left(\bigcup_{j \in J^c} A_j \right) = \bigcup_{i \in \mathbb{N}} A_i = \Omega.$$

Thus $\bigcup_{i \in J} A_i$ and $\bigcup_{j \in J^c} A_j$ partition Ω , so

$$B^c = \Omega \setminus B = \Omega \setminus \bigcup_{i \in J} A_i = \bigcup_{j \in J^c} A_j \in \mathcal{F},$$

since $J^c \subset \mathbb{N}$.

(d) Let $\{B_k\}_{k \geq 1} \subset \mathcal{F}$ be given.

For each $k \geq 1$, $\exists J_k \subset \mathbb{N}$ such that $B_k = \bigcup_{i \in J_k} A_i$.

Let $J \equiv \bigcup_{k \geq 1} J_k \subset \mathbb{N}$.

It then follows that since $J \subset \mathbb{N}$,

$$\bigcup_{k \geq 1} B_k = \bigcup_{k \geq 1} \bigcup_{i \in J_k} A_i = \bigcup_{i \in \bigcup_{k \geq 1} J_k} A_i = \bigcup_{i \in J} A_i \in \mathcal{F}.$$

By (a), (b), (d) (which, as in Q5 (1.4), also give closure under pairwise unions (Definition 1.1.1(c)) automatically) $\mathcal{F}$ satisfies Definition 1.1.1 and Definition 1.1.2. Hence $\mathcal{F}$ is a σ -algebra. //

Q7: Problem A

Let Ω be an uncountable set (e.g. $\mathbb{R}$) and $\mathbb{C}$ be the collection of all the singleton sets (i.e. $\mathbb{C} = \{\{\omega\} : \omega \in \Omega\}$). Find the smallest σ -algebra generated by $\mathbb{C}$. Justify your answer.

Answer

Sketch: We show that $\mathcal{F}_0$ is a σ -algebra containing $\mathbb{C}$ (so $\sigma\langle\mathbb{C}\rangle \subset \mathcal{F}_0$), then show that every σ -algebra containing $\mathbb{C}$ must contain $\mathcal{F}_0$ (so $\mathcal{F}_0 \subset \sigma\langle\mathbb{C}\rangle$).

Proof.

Step 1: $\mathcal{F}_0$ is a σ -algebra.

(a) $\Omega^c = \emptyset$ is countable, so $\Omega \in \mathcal{F}_0$.

(b) Let $A \in \mathcal{F}_0$ be given. If A is countable, then $(A^c)^c = A$ is countable, so $A^c \in \mathcal{F}_0$. If A^c is countable, then $A^c \in \mathcal{F}_0$ directly. In either case $A^c \in \mathcal{F}_0$.

(d) Let $\{A_n\}_{n \geq 1} \subset \mathcal{F}_0$ be given. If every A_n is countable, then $\bigcup_{n \geq 1} A_n$ is a countable union of countable sets, hence countable, so $\bigcup_{n \geq 1} A_n \in \mathcal{F}_0$. Otherwise, $\exists n_0$ with $A_{n_0}^c$ countable; since

$$\left(\bigcup_{n \geq 1} A_n \right)^c = \bigcap_{n \geq 1} A_n^c \subset A_{n_0}^c,$$

and a subset of a countable set is countable, $\left(\bigcup_{n \geq 1} A_n \right)^c$ is countable, so $\bigcup_{n \geq 1} A_n \in \mathcal{F}_0$.

By (a), (b), (d) (which, as in Q5 (1.4), also give closure under pairwise unions (Definition 1.1.1(c)) automatically) $\mathcal{F}_0$ satisfies Definition 1.1.1 and Definition 1.1.2. Hence $\mathcal{F}_0$ is a σ -algebra.

Step 2: $\mathbb{C} \subset \mathcal{F}_0$, hence $\sigma\langle\mathbb{C}\rangle \subset \mathcal{F}_0$.

For any $\omega \in \Omega$, $\{\omega\}$ is finite, hence countable, so $\{\omega\} \in \mathcal{F}_0$. Thus $\mathbb{C} \subset \mathcal{F}_0$. Since, by Definition 1.1.3, $\sigma\langle\mathbb{C}\rangle$ is the smallest σ -algebra containing $\mathbb{C}$ (i.e., the intersection of all σ -algebras containing $\mathbb{C}$), and $\mathcal{F}_0$ is a σ -algebra containing $\mathbb{C}$ (Steps 1–2), $\sigma\langle\mathbb{C}\rangle \subset \mathcal{F}_0$.

Step 3: $\mathcal{F}_0 \subset \sigma\langle\mathbb{C}\rangle$.

Let $\mathcal{F}$ be an arbitrary σ -algebra with $\mathbb{C} \subset \mathcal{F}$; it suffices to show $\mathcal{F}_0 \subset \mathcal{F}$, since $\sigma\langle\mathbb{C}\rangle$ is the intersection of all such $\mathcal{F}$.

Let $A \in \mathcal{F}_0$ be given.

Case A countable: If $A = \emptyset$, then $A = \Omega^c \in \mathcal{F}$, since $\Omega \in \mathcal{F}$ and $\mathcal{F}$ is closed under complementation. If $A \neq \emptyset$, enumerate $A = \{\omega_1, \omega_2, \dots\}$ (finite or countably infinite). Since $\mathbb{C} \subset \mathcal{F}$, $\{\omega_i\} \in \mathcal{F}$ for every i , and since $\mathcal{F}$ is a σ -algebra (closed under countable unions), $A = \bigcup_i \{\omega_i\} \in \mathcal{F}$.

Case A^c countable: By the previous case applied to A^c , $A^c \in \mathcal{F}$; since $\mathcal{F}$ is closed under complementation, $A = (A^c)^c \in \mathcal{F}$.

In either case $A \in \mathcal{F}$. Since $A \in \mathcal{F}_0$ was arbitrary, $\mathcal{F}_0 \subset \mathcal{F}$. As $\mathcal{F}$ was an arbitrary σ -algebra containing $\mathbb{C}$, $\mathcal{F}_0 \subset \mathcal{F}$ for every such $\mathcal{F}$, hence $\mathcal{F}_0 \subset \bigcap \mathcal{F} = \sigma\langle \mathbb{C} \rangle$.

By Steps 2-3, $\sigma\langle \mathbb{C} \rangle = \mathcal{F}_0 = \{A \subset \Omega : A \text{ or } A^c \text{ is countable}\}.$